

第一大题:填空题(18')

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填空题_2:求输入电阻 R_{in} (3')

填空题_3:求某一节点的自导(3')

填空题_4:电路响应的判别(3')

填空题_5:复功率($\tilde{S}=P+jQ$)(3')

填空题_6:判断某一端口容性/感性(3')

第二大题:选择题(15')

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选择题_2:零输入响应(3')

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计算题_1:KVL,KCL 的运用(7')

计算题_2:回路电流法求 U 和 I(10')

计算题_3:求 $I_1, 1/\omega C, \omega L, R_2$ (10')

计算题_4:全响应(15')

计算题_5:最大功率问题(15')

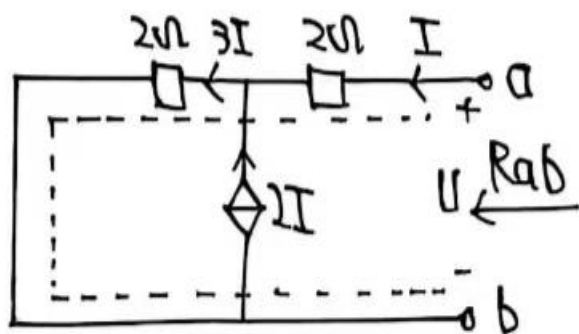
第四大题:功率,功率因数(10')

填空题_1:两点之间电压降的求法(3')

电路中任意两点之间的电压降的求法:

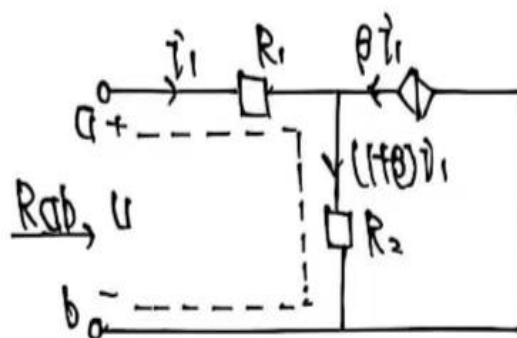
1. 确定总参考方向;
2. 看每个元件上面的各自的参考方向是否与总的电压降参考方向是否一致,
一致正号,不一致负号,带具体数值;
3. 任意两点之间的电压降的求法与所走的路径无关。

填空题_2:求输入电阻 $R_{in}(3')$



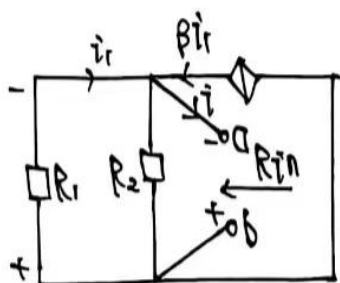
$$U = 2I + 2 \times 3I = 8I$$

$$\therefore R_{ab} = \frac{U}{I} = 8\Omega$$



$$U = R_1 i_1 + R_2 (1 + \beta) i_1$$

$$R_{ab} = \frac{U}{i_1} = R_1 + (1 + \beta) R_2$$



$$R_1 = R_2 = 100\Omega$$

$$\beta = 18$$

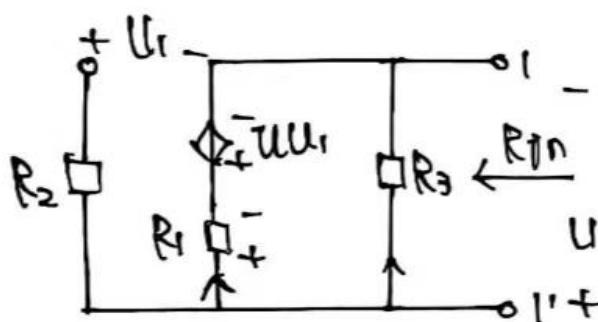
$$R_{in} = 5\Omega$$

ab之间可能还有其它元件

$$i = i_1 + \beta i_1 + \frac{i_1 R_1}{R_2}$$

$$R_{in} = \frac{U}{I} = \frac{i_1 R_1}{i_1 + \beta i_1 + \frac{i_1 R_1}{R_2}}$$

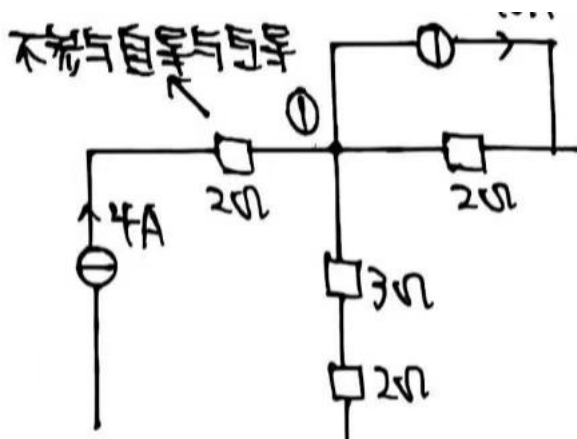
$$= \frac{R_1 R_2}{(1 + \beta) R_2 + R_1}$$



$$R_{in} = \frac{U}{I} = \frac{U_1}{\frac{U_1}{R_3} + \frac{U_1 - U}{R_1}}$$

$$= \frac{1}{\frac{1}{R_3} + \frac{1 - U}{R_1}}$$

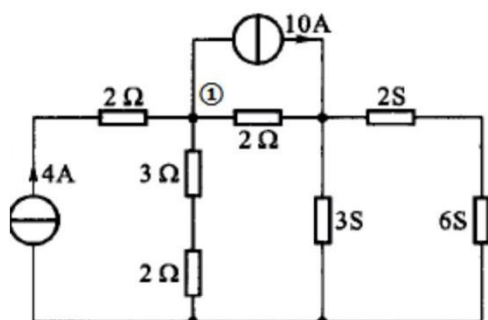
填空题_3:求某一节点的自导(3')



$$1-1$$

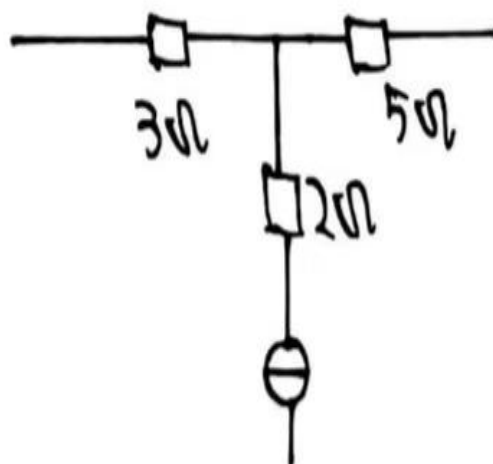
$$\left(\frac{1}{2} + \frac{1}{2+3}\right) = 0.7$$

5、**单选** 图式电路中在列写结点电压方程时。图中结点1的自导是 () S。



(5分)

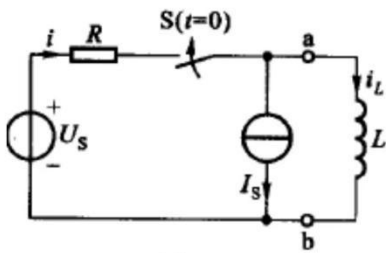
- ☐ A 0.5
- ☐ B 0.2
- ☐ C 1.5
- ☒ 0.7



$$\left(\frac{1}{3} + \frac{1}{5}\right) = \frac{8}{15}$$

填空题_4:电路响应的判别(3')

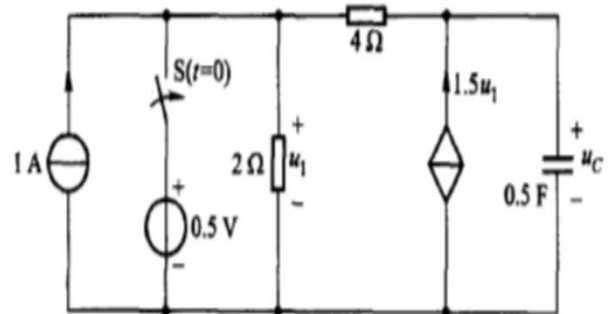
5、单选 图示电路假设开关在 $t=0$ 时刻发生动作。试问该电路中的电感电流 $i_L(t)$ 属于()响应。



(5分)

- ☐ A 零状态
- ☒ B 全
- ☐ C 零输入
- ☐ D 阶跃或冲激

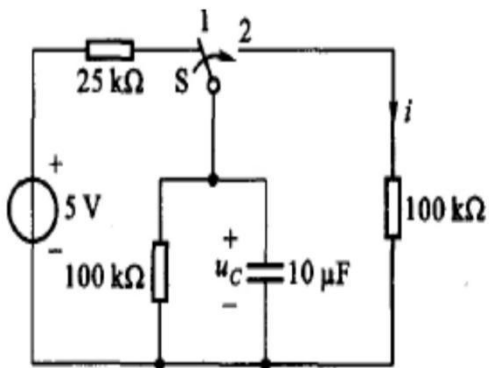
6、填空 试判断图示电路属于什么响应? ()



(5分)

全

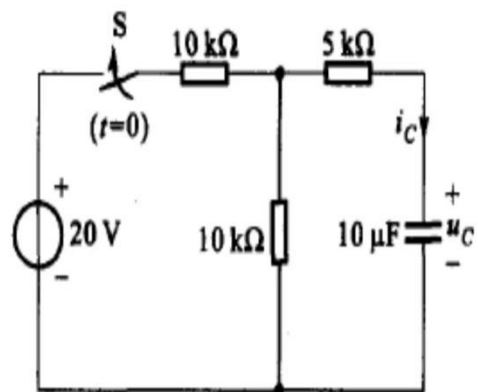
7、填空 试判断图示电路属于什么响应? ()



(5分)

零输入

8、填空 试判断图示电路属于什么响应? ()
假设电容C初始能量为0.



(5分)

零状态

填空题_5:复功率($\tilde{S}=P+jQ$)(3')

例题 1: 假设线性电路中某条支路的有功功率 $P=20W$,无功功率 $Q=15VAR$,

试问视在功率 S 等于(25)VA,复功率 $\tilde{S}=P+jQ=20+j15$ (VA)

例题 2: 已知 $P=60 KW$, 功率因数 $=0.6$, 则 $S=100 KVA$

$Q = 100 * 0.8 = 80 K VAR$, 复功率 $\tilde{S}=60+j80 KVA$

填空题_6:判断某一端口容性/感性(3')

6、☒ 填空 已知某一段电路的电压和电流函数如下。试问u和i的相位关系是 ()

$$u = 10 \sin(10^3 t - 20^\circ) \text{ V}$$

$$i = 2 \cos(10^3 t - 50^\circ) \text{ A}$$

(5分)

u 滞后 i 60°

$$u = 10 \sin(10^3 t - 20^\circ) = 10 \cos(10^3 t - 110^\circ)$$

$$U = 5\sqrt{2} \angle -110^\circ \quad I = \sqrt{2} \angle -50^\circ$$

$$\varphi_u - \varphi_i = -60^\circ \text{ 故 } i \text{ 超前 } u \ 60^\circ$$

容性

$$\sin x = \cos(x - \frac{\pi}{2})$$

8、☒ 填空 已知某支路的电压、电流(关联参考方向)表达式如下。试问该支路可能会 () 串联或并联。

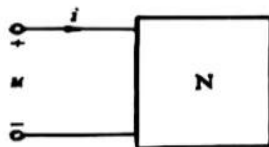
$$\begin{cases} u = 10 \cos(314 t + 45^\circ) \text{ V} \\ i = 2 \cos(314 t) \text{ A} \end{cases}$$

(5分)

R 与 L 或者 R L C

2、☒ 填空 图中N为不含独立源的一端口。端口处的电压和电流如下。

$u = 100 \cos(2t + 60^\circ) \text{ V}$, $i = 5 \cos(2t - 30^\circ) \text{ A}$ 。试求端口处的导纳Y等于 ()。



(5分)

A $20j$

B $10j$

☒ C $1/(20j)$

D $1/(10j)$

$$u = 100 \cos(\omega t - 30^\circ)$$

$$i = -2 \sin(\omega t + 75^\circ)$$

$$= 2 \cos(\omega t + 165^\circ)$$

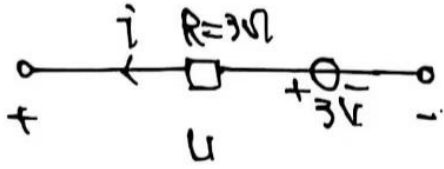
u 超前 i 165°

感性。

$$-\sin x = \cos(x + \frac{\pi}{2})$$

选择题_1: U 与 I 之间的约束关系(3')

例:

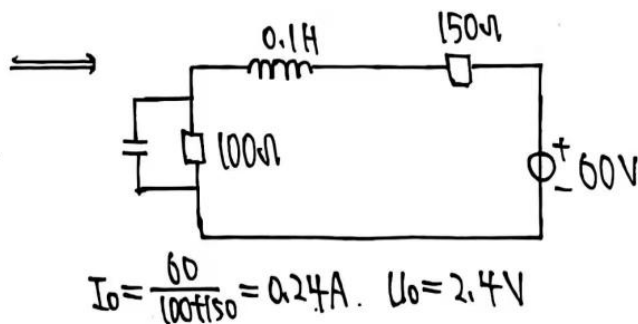
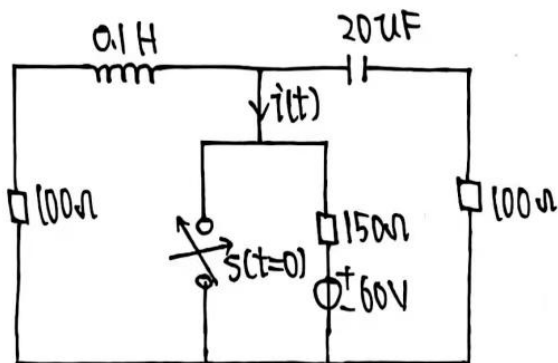


故 $U = -3i + 3$

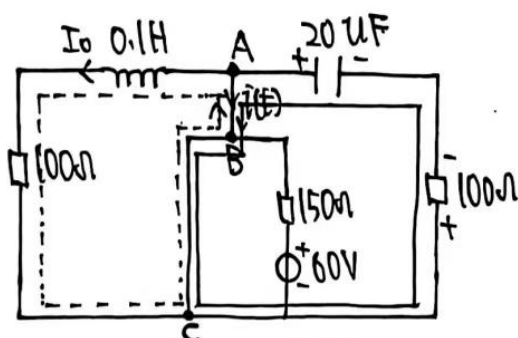
注意: U 与 I 的参考方向

选择题_2: 零输入响应(3')

⑧ S在 $t=0$ 时合上,求S闭合后, $i(t)=?$



S闭合以后.



若求BC段的电流,还有直流独立电压源
60V单独作用产生的电流0.4A

$$i_{B \rightarrow C}(t) = 0.24(e^{-500t} - e^{-1000t}) + 0.4$$

L: 用虚线表示电感作用的回路:

$$i_L(t) = I_0 \cdot e^{-\frac{t}{\tau}}$$

$$\tau = \frac{L}{R_{eq1}} = \frac{0.1}{100\Omega} = 1ms$$

$$i_L(t) = 0.24 \cdot e^{-1000t}$$

$$i_L(t) \text{ 从 } B \rightarrow A$$

C: 用实线表示电感C放电的回路

$$U_C(t) = U_0 \cdot e^{-\frac{t}{\tau_2}}$$

$$\tau_2 = R_{eq2} \cdot C = 20 \times 10^{-6} \times 100 = 2ms$$

$$U_C(t) = 2.4 \cdot e^{-500t}$$

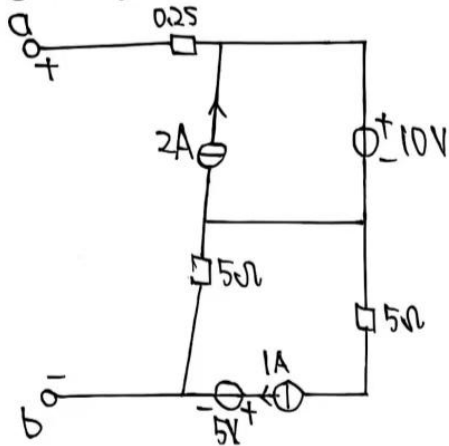
$$i_C(t) = -C \frac{dU_C(t)}{dt} = 0.24 e^{-500t}$$

$$i_C(t) \text{ 的方向 } A \rightarrow B$$

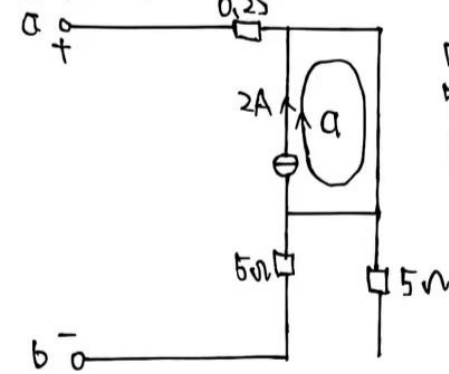
$$\text{故 } i(t) = i_C(t) - i_L(t) = 0.24(e^{-500t} - e^{-1000t}).$$

选择题_3:叠加定理(3')

④ 求 U_{ab} :



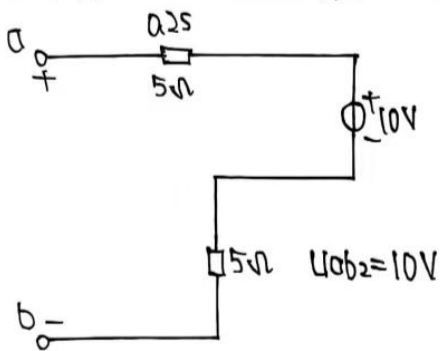
a: 只有 2A 的独立电流源单独作用



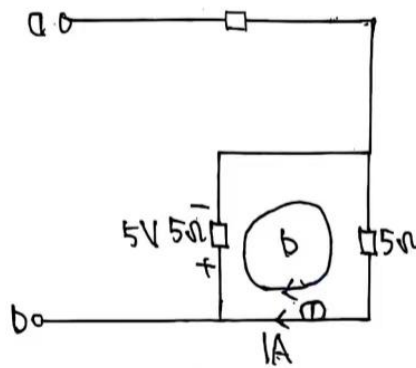
由于 2A 的独立电流源被短路, 故其 2A 电流只在 a 回路中流。

$$U_{ab1} = 0V$$

b: 只有 10V 的独立电压源单独作用



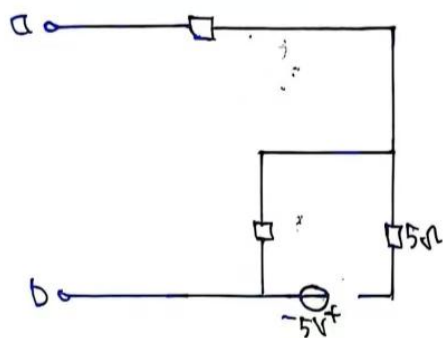
c: 只有 1A 的独立电流源单独作用



由于与其他构不成闭合回路, 故 1A 只在回路 c 里面流。

$$\therefore U_{ab3} = -5V.$$

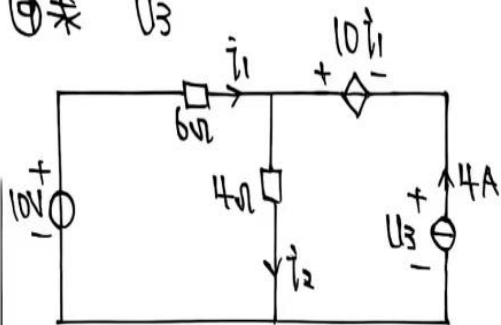
d: 当只有 5V 的独立电压源单独作用:



$$U_{ab4} = 0V$$

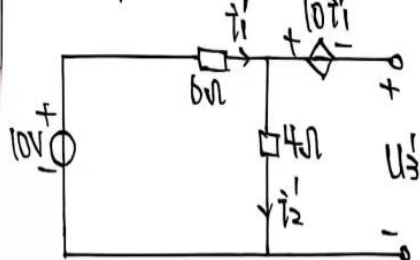
$$\text{故 } U_{ab} = U_{ab1} + U_{ab2} + U_{ab3} + U_{ab4} = 5V$$

求 U_3



$$U_3 = U_3' + U_3'' = 19.6V$$

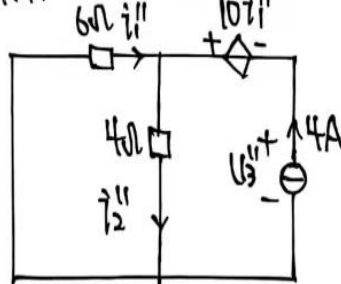
a: 只有电压源 10V 单独作用



$$i_1' = i_2' = 1A$$

$$U_3' = -10i_1' + 4i_2' = -6V$$

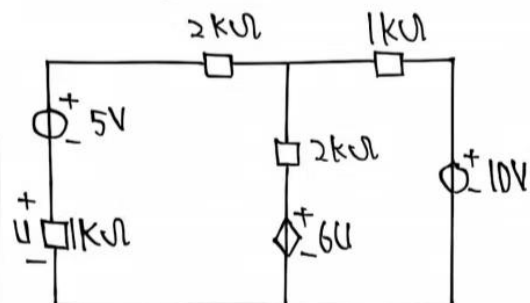
b: 只有电流源 4A 单独作用



$$i_2'' = \frac{6}{6+4} \times 4 = 2.4A, i_1'' = -1.6A$$

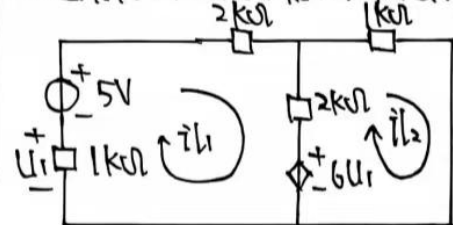
$$U_3'' = -10i_1'' + 4i_2'' = 16 + 9.6 = 25.6V$$

4-4 (2) 叠加定理:



$$U = U_1 + U_2 = 1V$$

a: 当只有 5V 独立电压源单独作用.



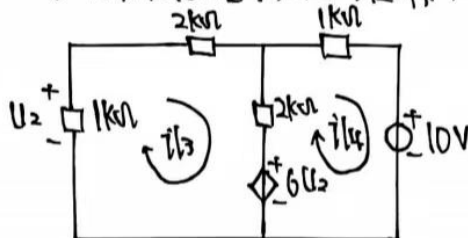
$$\text{回1: } 5000i_1 - 2000i_2 = 5 - 6U_1$$

$$\text{回2: } -2000i_2 + 3000i_2 = 6U_1$$

$$\text{补充: } U_1 = -1000i_1$$

$$\text{解得: } \begin{cases} i_1 = 3mA \\ i_2 = -4mA \\ U_1 = -3V \end{cases}$$

b: 当只有 10V 的独立电压源单独作用.



$$\text{回3: } 5000i_3 - 2000i_4 = -6U_2$$

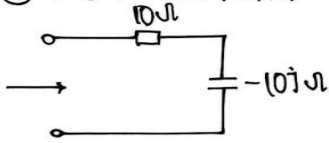
$$\text{回4: } -2000i_3 + 3000i_4 = 6U_2 - 10$$

$$\text{补充: } U_2 = -1000i_3$$

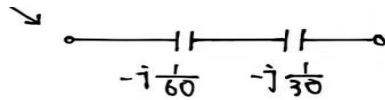
$$\text{解得: } \begin{cases} i_3 = -4mA \\ i_4 = 2mA \\ U_2 = 4V \end{cases}$$

选择题_4:电表的读数(3')

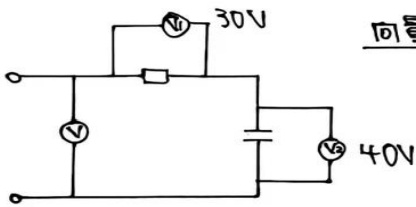
⑧ 电感与电阻串联.



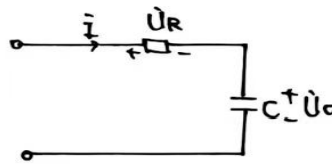
$$R = 10 - j10$$



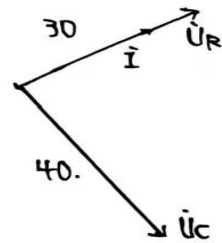
$$-j\frac{1}{60} - j\frac{1}{30} = -j\frac{1}{20}$$



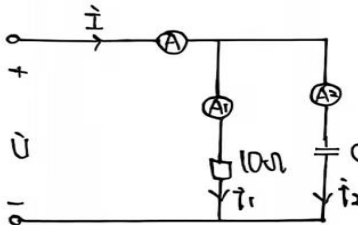
向量表示



$$\therefore V = 50V$$



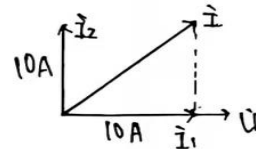
⑨ 电阻与电容并联.



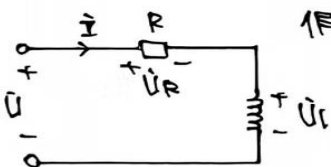
$$A_1 = 10A$$

$$A_2 = 10A$$

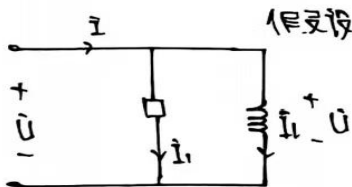
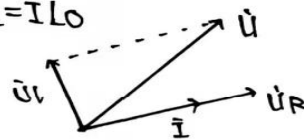
$$A = 10\sqrt{2}A$$



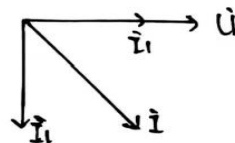
⑩ 电感与电阻串联与并联.



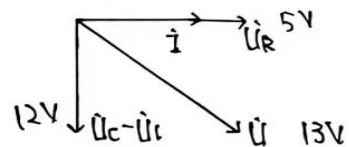
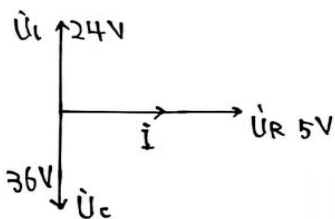
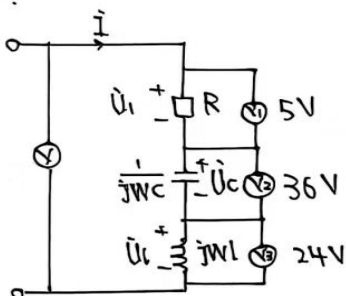
假设 $i = I \angle 0^\circ$



假设 $U = U \angle 0^\circ$

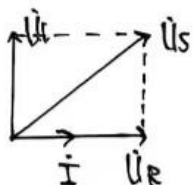
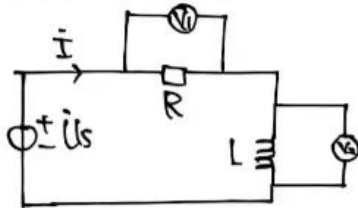


⑪:



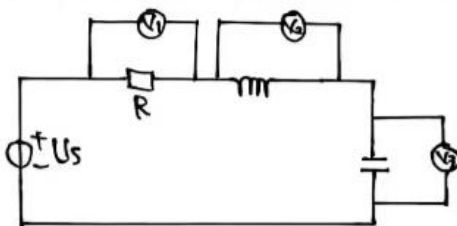
8-10:

a: 已知 $V_1: 30V, V_2: 60V$



故 $U_s = 30\sqrt{5} (V)$

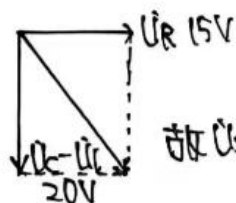
b: 已知 $V_1: 15V, V_2: 80V, V_3: 100V$



$U_L 80V$

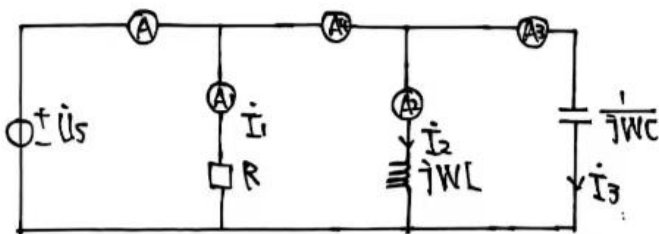
$U_R 15V$

$U_C 100V$



故 $U_s = 25V$

8-5: 已知: $A_1: 5A, A_2: 20A, A_3: 25A$, 求 A 和 A_4 的读数.



设 $U_s = U_s \angle 0^\circ (V)$

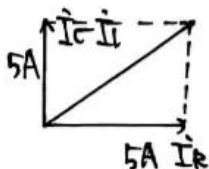
$I_C 25A$

$U_s = U_L = U_C$

$I_R: 5A$

$I_L 20A$

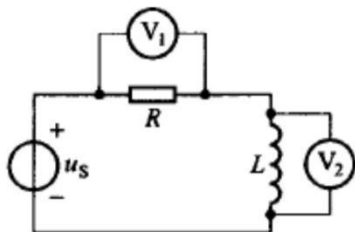
故 $A_4: I_C - I_L = 5A$



故 $A: 5\sqrt{2}A$

表 $A: 7.07A$, 表 $A_4: 5A$.

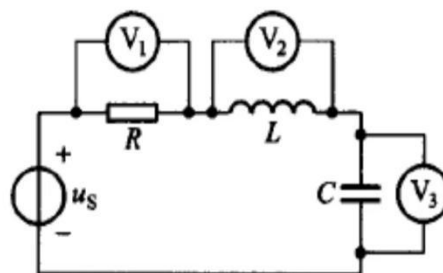
5、**单选** 已知图示电路中的 V_1 和 V_2 的读数分别是30V和60V。试问图中电压源 u_s 的有效值 U_s 大小约等于 () V。



(5分)

- ☐ A 90
- ☐ B 30
- ☒ C 67.08
- ☐ D 60

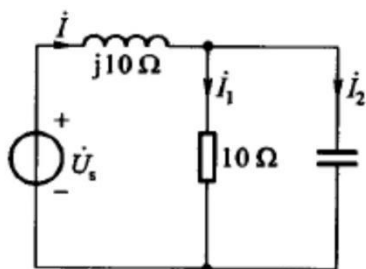
7、**填空** 图示电路的中 V_1 、 V_2 、 V_3 读数分别是15V、80V、100V。现问图中电源 u_s 的有效值 U_s 等于 () V。



(5分)

25

9、**填空** 图示电路中的 $I_1=I_2=10A$ 。试问图中的 I 大小约等于 () A。

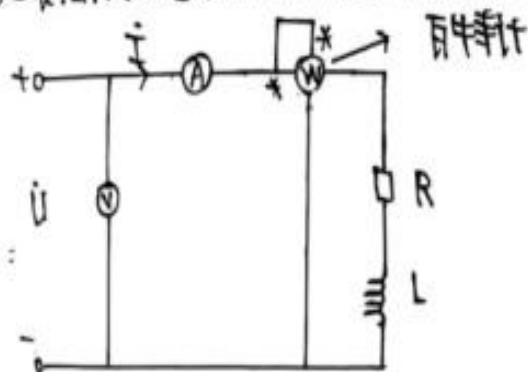


(5分)

14.14

选择题_5:三表法测线圈(3')

②三表法测线圈(线圈有内阻等效有R和L)



已知: $U=50V$, $I=1A$,
 $f=50Hz$, $P=30W$, 求 $L=?$, $R=?$

$$P=30W=I^2R=1^2 \times R \Rightarrow R=30\Omega$$

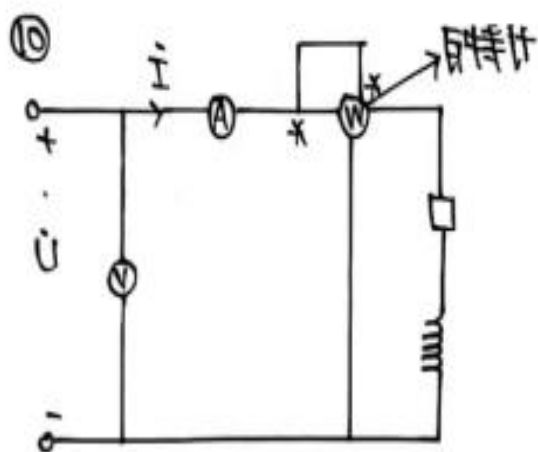
$$|Z|=\sqrt{R^2+(WL)^2}=\frac{U}{I}=50$$

$$\Rightarrow WL=40\Omega$$

$$L=\frac{40}{2\pi f}=\frac{40}{2\pi \times 50}=\frac{0.4}{\pi}=0.127H$$

$$W=\frac{2\pi}{T}=2\pi f = \frac{0.4}{\pi} \times 10^3 \text{ mH} = 127.32 \text{ mH}$$

更精确一点



已知: V读数: $50V$, A读数: $1A$.
 $W=1000 \text{ rad/s}$, W读数: $30W$,
 求线圈 L 和 R 的大小

$$\frac{U}{I}=R+jWL \quad W=1000 \text{ rad/s}$$

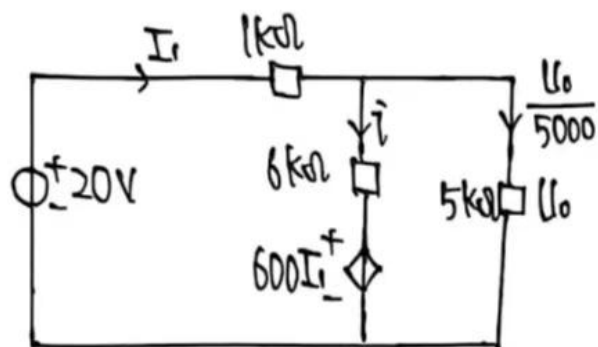
$$\frac{U}{I}=\sqrt{R^2+(WL)^2}=\frac{50}{1}=50\Omega$$

$$P=30=I^2R=1 \times R \Rightarrow R=30\Omega$$

$$\text{故 } WL=40\Omega \rightarrow L=40 \text{ mH}$$

计算题_1: KVL,KCL 的运用(7')

1-18: 求 I_1 和 U_0

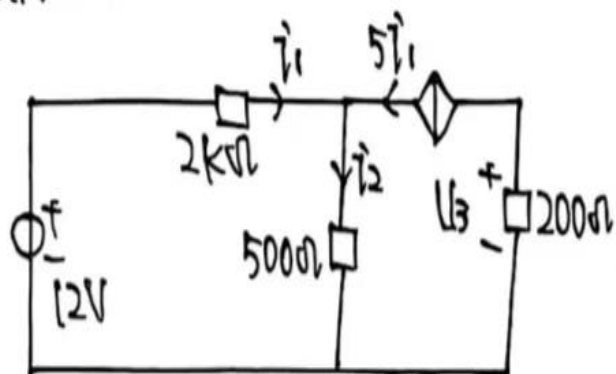


$$\text{KCL: } I_1 = i + \frac{U_0}{5000}$$

$$\text{KVL: } \begin{cases} 20\text{V} = 1000I_1 + U_0 \\ U_0 = 6000i + 600I_1 \end{cases}$$

$$\text{解得: } \begin{cases} U_0 = 15\text{V} \\ I_1 = 5\text{mA} \end{cases}$$

1-14:



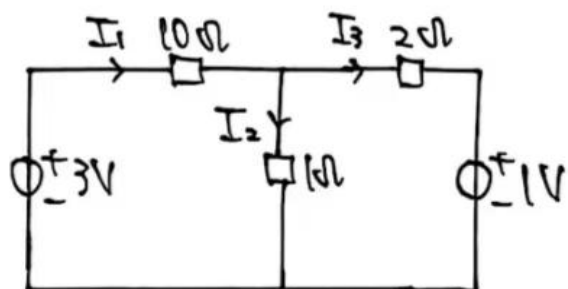
$$\text{KCL: } i_2 = i_1 + 5i_1 = 6i_1$$

$$\text{KVL: } 12\text{V} = 2000i_1 + 500i_2$$

$$\text{解得: } i_1 = 2.4\text{mA}$$

$$U_3 = -200 \times 5i_1 = -2.4\text{V}$$

1-3:



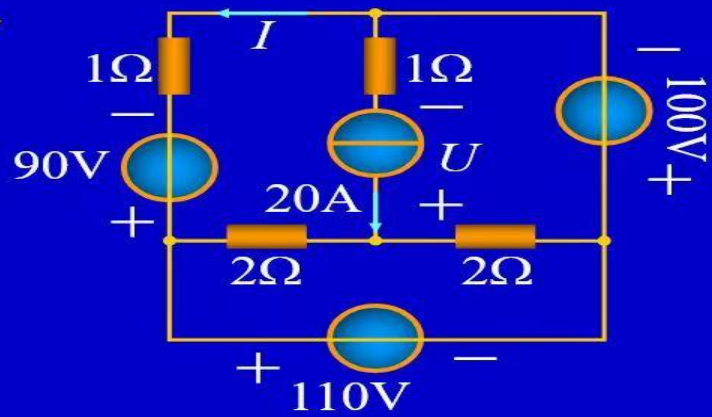
$$\text{KCL: } I_1 = I_2 + I_3$$

$$\text{KVL: } \begin{cases} 3\text{V} = 10I_1 + I_2 \\ I_2 = 2I_3 + 1 \end{cases}$$

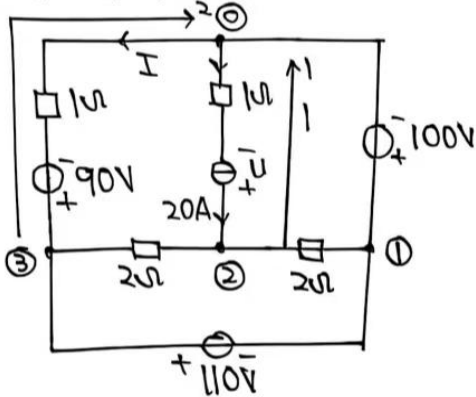
$$\text{解得: } \begin{cases} I_1 = 0.25\text{A} \\ I_2 = 0.5\text{A} \\ I_3 = -0.25\text{A} \end{cases}$$

计算题 2: 用回路电流法求 U 和 $I(10')$

求电压 U 和电流 I 。



① 求电压 U 和电流 I 。



由箭头1: $V_2 - V_0 = U - 20$, 故 $U = 195V$

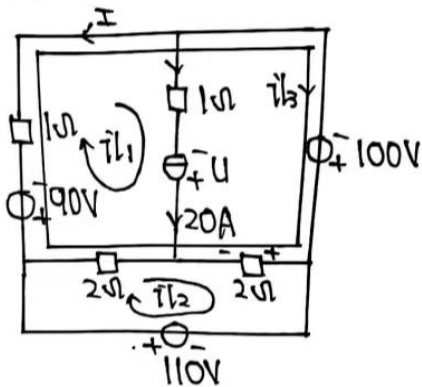
由箭头2: $V_3 - V_0 = 90V - I$, 故 $I = -120A$.

a: 用超级节点法处理。

$$\begin{cases} \text{结点①: } V_1 = 100V \\ \text{结点③: } V_3 = V_1 + 110V = 210V \\ \text{结点②: } -\frac{1}{2}U_{11} + (\frac{1}{2} + \frac{1}{2})U_{12} - \frac{1}{2}U_{13} = 20. \end{cases}$$

解得 $\begin{cases} V_1 = 100V \\ V_2 = 175V \\ V_3 = 210V. \end{cases}$

b: 用回路电流法:



回路1: $i_{l1} = 20A$

回路2: $-2i_{l1} + 4i_{l2} - 4i_{l3} = 110V$

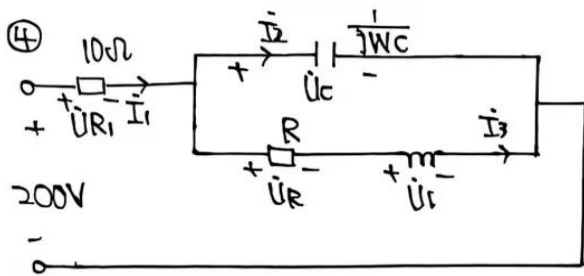
回路3: $3i_{l1} - 4i_{l2} + 5i_{l3} = 100 - 90$

解得 $\begin{cases} i_{l1} = 20A \\ i_{l2} = 137.5A \\ i_{l3} = 100A \end{cases}$

$I = -i_{l1} - i_{l3} = -120A$

$100V = U - 20 + 2(i_{l3} - i_{l2}), U = 195V.$

计算题 3: 求 $I_1, 1/\omega C, \omega L, R_2(15')$



$I_1 = I_2 + I_3$. 设 $\omega L = R$.

已知 $I_2 = 10A$, $I_3 = 10\sqrt{2}A$

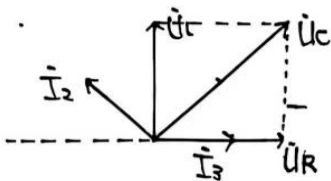
设 $I_3 = 10\sqrt{2} \angle 0^\circ A$

$\dot{U}_R = R \times 10\sqrt{2} \angle 0^\circ$

$\dot{U}_L = j\omega L \times 10\sqrt{2} \angle 0^\circ$

$\dot{U}_L = \dot{U}_R + \dot{U}_C = \dot{U}_C = U_C \angle 45^\circ$

并联的两部分 $\dot{U}$ (电压) 相等



由 $\dot{U}_C = U_C \angle 45^\circ \longrightarrow I_2 = 10 \angle 135^\circ$

$\therefore \dot{I}_1 = \dot{I}_2 + \dot{I}_3 = -5\sqrt{2} + 5\sqrt{2}j + 10\sqrt{2} = 5\sqrt{2} + 5\sqrt{2}j$
 $= 10 \angle 45^\circ + 10\sqrt{2} \angle 0^\circ = 10 \angle 45^\circ$

$\dot{I}_1$ 与 $\dot{U}_C$ 同相. $\dot{U}_R = \dot{I}_1 \times 10\Omega = 100 \angle 45^\circ$ 有效值 100V

$U_C = 200 - 100 = 100V$, 即 $\dot{U}_C = 100 \angle 45^\circ$

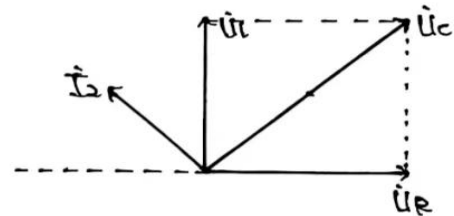
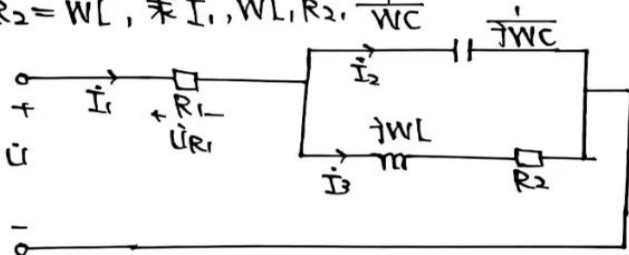
$\dot{U}_C = \dot{I}_2 \times \frac{1}{j\omega C} = 10 \angle 135^\circ \times \frac{1}{\omega C} \angle -90^\circ = 10 \times \frac{1}{\omega C} \angle 45^\circ$, $\omega C = \frac{1}{10} = 0.1$, $\frac{1}{\omega C} = 10\Omega$

$\times \dot{U}_C = \dot{U}_R + \dot{U}_L$, 故 $U_C = U_R = 50\sqrt{2}V$. $U_R = I_3 \cdot R = 10\sqrt{2} \cdot R$, 故 $\omega L = R = 5\Omega$.

⑩ 图不中足各:

$I_2 = 10A$, $I_3 = 10\sqrt{2}A$, $U = 200V$, $R_1 = 5\Omega$

$R_2 = \omega L$, 求 $I_1, \omega L, R_2, \frac{1}{\omega C}$



解: 设 $I_3 = 10\sqrt{2} \angle 0^\circ A$

$U_R = R_2 \times 10\sqrt{2} \angle 0^\circ (V)$ $\dot{U}_L = 10\sqrt{2} \angle 0^\circ \times \omega L \angle 90^\circ = \omega L \times 10\sqrt{2} \angle 90^\circ$

设 $R_2 = \omega L$, 故 $|\dot{U}_R| = |\dot{U}_L|$

$\dot{U}_C = \dot{U}_R + \dot{U}_L = U_C \angle 45^\circ \longrightarrow I_2$ 超前 $\dot{U}_C$ 90° ,

故 $\dot{I}_1 = \dot{I}_2 + \dot{I}_3 = 10 \angle 135^\circ + 10\sqrt{2} \angle 0^\circ = 5\sqrt{2} + 5\sqrt{2}j = 10 \angle 45^\circ A$

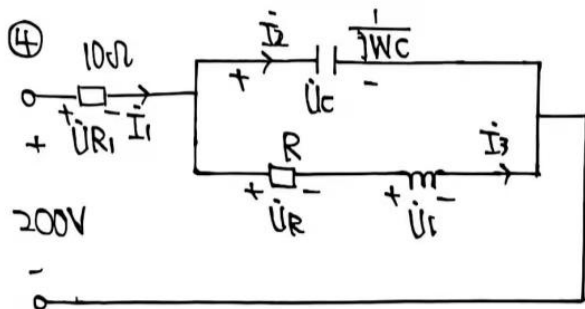
$\dot{U}_R = 5 \dot{I}_1 = 50 \angle 45^\circ V$.

$\dot{U}_R$ 与 $\dot{U}_C$ 同相位. $U = 200 \angle 45^\circ = \dot{U}_R + \dot{U}_C = 200 \angle 45^\circ$.

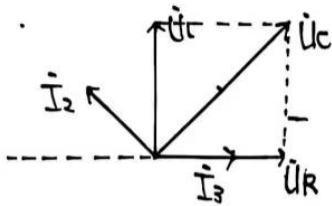
$200 = U_R + U_C$, $U_C = 200 - 50 = 150V$

$U_C = \sqrt{U_R^2 + U_L^2} = \sqrt{2} (R_2 \times 10\sqrt{2}) = 150$ 解得 $\omega L = R_2 = 7.5\Omega$

$I_2 = 10A$, $\frac{1}{\omega C} = \frac{U_C}{I_2} = \frac{150}{10} = 15\Omega$



$$I_1 = I_2 + I_3, \text{ 设 } \omega L = R.$$



已知 $I_2 = 10A, I_3 = 10\sqrt{2}A$

设 $I_3 = 10\sqrt{2} \angle 0^\circ A$

$$\dot{U}_R = R \times 10\sqrt{2} \angle 0^\circ$$

$$\dot{U}_L = j\omega L \times 10\sqrt{2} \angle 0^\circ$$

$$\dot{U}_L = \dot{U}_R + \dot{U}_C = \dot{U}_C = U_C \angle 45^\circ$$

并联的两部分 $\dot{U}$ (电压量) 相等

由 $\dot{U}_C = U_C \angle 45^\circ \longrightarrow I_2 = 10 \angle 135^\circ$

$$\begin{aligned} \therefore \dot{I}_1 &= \dot{I}_2 + \dot{I}_3 = -5\sqrt{2} + 5\sqrt{2}j + 10\sqrt{2} = 5\sqrt{2} + 5\sqrt{2}j \\ &= 10 \angle 135^\circ + 10\sqrt{2} \angle 0^\circ = 10 \angle 45^\circ \end{aligned}$$

$\dot{I}_1$ 与 $\dot{U}_C$ 同相. $\dot{U}_R = \dot{I}_1 \times 10\Omega = 100 \angle 45^\circ$ 有效值 100V

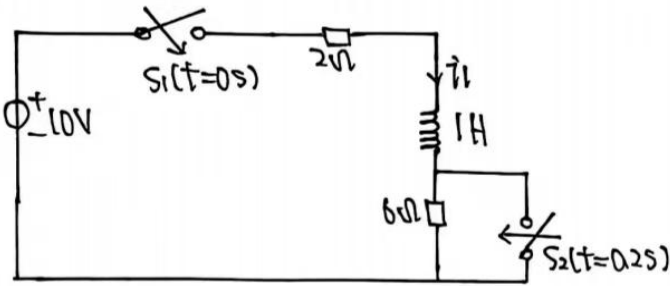
$U_C = 200 - 100 = 100V$, 即 $U_C = 100$

$$\dot{U}_C = \dot{I}_2 \times \frac{1}{j\omega C} = 10 \angle 135^\circ \times \frac{1}{\omega C} \angle -90^\circ = 10 \times \frac{1}{\omega C} \angle 45^\circ, \quad \omega C = \frac{1}{10} \therefore 0.1 \quad \frac{1}{\omega C} = 10\Omega$$

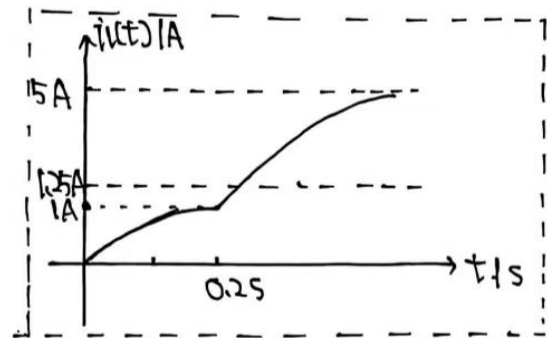
$\times \dot{U}_C = \dot{U}_R + \dot{U}_L$, 故 $U_C = U_R = 50\sqrt{2}V$ $\therefore U_R = I_3 \cdot R = 10\sqrt{2} \cdot R$, 故 $\omega L = R = 5\Omega$.

计算题 4:全响应(15')

⑧ 求两次切换后的 $i_L(t)$

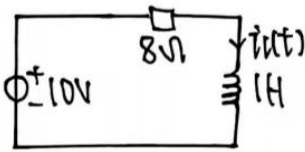


画的不好以佳.



$$i_L(0^-) = i_L(0^+) = 0A.$$

a: $0 \rightarrow 0.25$ 时, 零状态

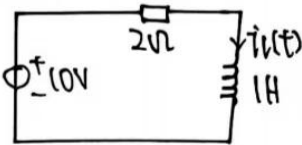


$$i_L(t) = I_s(1 - e^{-\frac{t}{\tau_1}}) = \frac{10}{8}(1 - e^{-\frac{t}{\frac{1}{8}}})$$

$$= 1.25(1 - e^{-8t}) (A).$$

$$i_L(t)|_{t=0.2} = 1.25 \times (1 - \frac{1}{e^{1.6}}) = 1.25(1 - 0.2) = 1A.$$

b: $t > 0.25$, 全响应



$$i_L(0.2^+) = 1A,$$

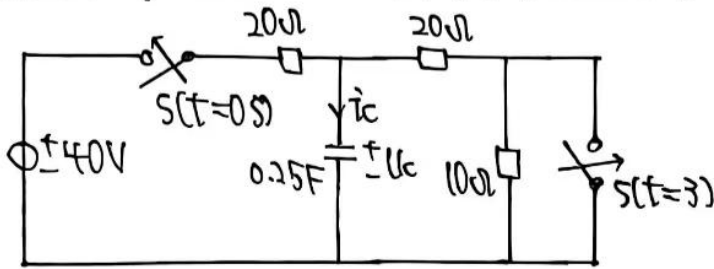
$$i_L(\infty) = \frac{10}{2} = 5A, \tau_2 = \frac{L}{R} = \frac{1}{2}s$$

$$i_L(t) = i_L(\infty) + [i_L(0.2^+) - i_L(\infty)] \cdot e^{-\frac{t-0.2}{\tau_2}}$$

$$= 5 + (1-5)e^{-2(t-0.2)}.$$

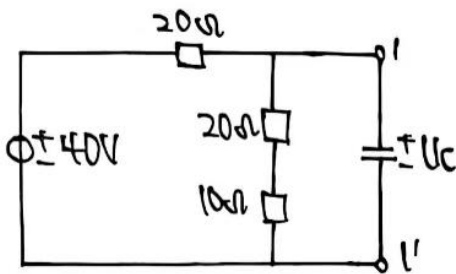
$$= 5 - 4 \cdot e^{-2(t-0.2)} (A)$$

例7-5: 已知 $U_C(0^-) = 0$, 求 $t > 0$ 后, $U_C(t)$, $i_C(t)$



在 $0^- \sim 0^+$ 期间, $U_C(0^+) = U_C(0^-) = 0V$.

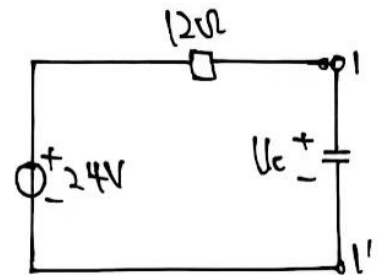
在 $0 < t < 3s$ 时, C: 充电状态.



1-1' 以左戴维南等效

$$U_{OC} = \frac{20 \parallel 10}{20 \parallel 10 + 20} \times 40 = 24V$$

$$R_{eq} = \frac{30 \times 20}{30 + 20} = 12\Omega$$



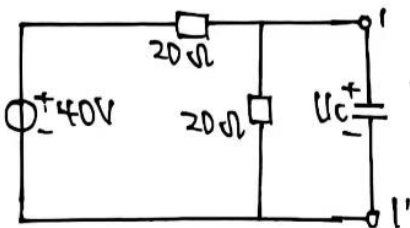
$$U_{OC} = 24V, \quad T = R_{eq} \times C = 12 \times 0.25 = 3s$$

$$\text{故 } U_C(t) = U_S(1 - e^{-\frac{t}{T}}) = 24(1 - e^{-\frac{t}{3}}) (V)$$

$$i_C(t) = C \cdot \frac{dU_C(t)}{dt} = -24 \times (-\frac{1}{3}) \times 0.25 \cdot e^{-\frac{t}{3}} = 2e^{-\frac{t}{3}} (A)$$

$$t=3s \text{ 时 } U_C(t)|_{t=3} = 24(1 - e^{-1}) = 15.17V.$$

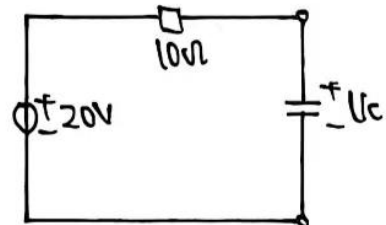
$$\text{在 } t > 3s \text{ 时, C: 放电状态: } U_C(t) = U_C(\infty) + [U_C(3s) - U_C(\infty)] \cdot e^{-\frac{t-3}{T}}$$



1-1' 以左戴维南等效

$$U_{OC} = \frac{20}{20 + 20} \times 40 = 20V$$

$$R_{eq2} = 20 \parallel 20 = 10\Omega$$



$$U_C(t) = U_C(\infty) + [U_C(3s) - U_C(\infty)] \cdot e^{-\frac{t-3}{T}}$$

$$U_C(\infty) = 20V, \quad U_C(3s) = 15.17V, \quad T = R_{eq2} C = 10 \times 0.25 = 2.5s$$

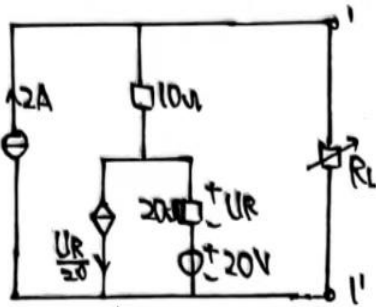
$$U_C(t) = U_C(\infty) + [U_C(3s) - U_C(\infty)] \cdot e^{-\frac{t-3}{T}} = 20 + [15.17 - 20] \cdot e^{-\frac{t-3}{2.5}} (V)$$

$$= 20 - 4.83 \cdot e^{-0.4(t-3)} (V)$$

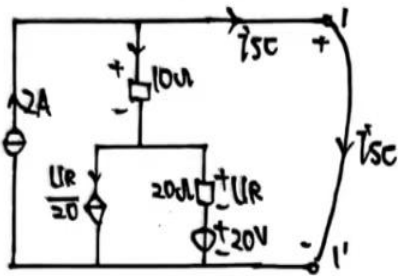
$$i_C(t) = C \frac{dU_C(t)}{dt} = -4.83 \times (-0.4) \times 0.25 \cdot e^{-0.4(t-3)} = 0.483 \cdot e^{-0.4(t-3)} (A)$$

计算题 5:最大功率(15')

⑧求 R_L 上的 $P_{max}=?$

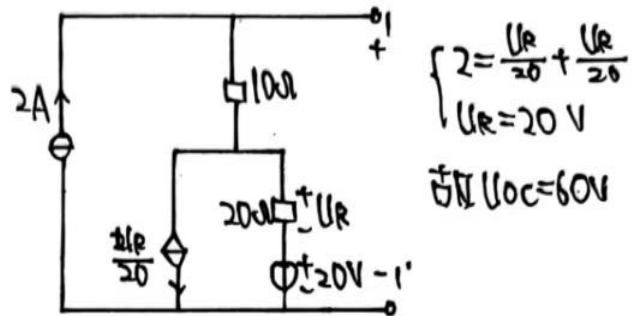


b: 求 I_{sc} , 短路电流:

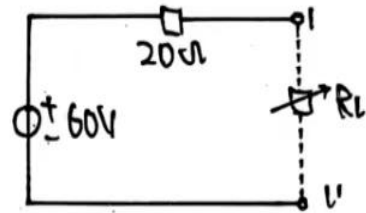


c: 求 $R_{eq} = \frac{U_{oc}}{I_{sc}} = 20\Omega$

a: 求 U_{oc} , 1-1'端的开路电压



故1-1'端的戴维南等效



$$P_{max} = \frac{U_{oc}^2}{4R_{eq}} = 45W$$

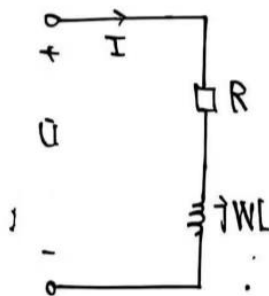
$$\begin{cases} 2A = 2 \times \frac{U_R}{20} + I_{sc} \\ 10 \times \frac{U_R}{10} + U_R + 20 = 0 \end{cases}$$

故 $\begin{cases} U_R = -10V \\ I_{sc} = 3A \end{cases}$

第四大题:功率,功率因数(10')

③ 已知 $f=50\text{Hz}$, $U=380\text{V}$, $\lambda_1=0.6$, $P_1=20\text{kW}$.

若 $\lambda_1=0.6$ 变为 $\lambda_2=0.9$

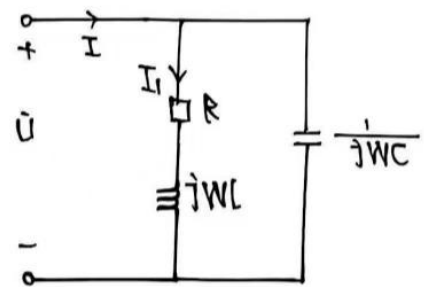


$$P = UI \cos(\varphi_u - \varphi_i) = UI \cdot \lambda$$

λ 变大 $\rightarrow Q$ 小, $\sin(\varphi_u - \varphi_i)$ 小.

$Q_C < 0$, 可以通过并联一个 C

来减小 Q , 这个过程中 P 不变



$$(1) P_1=20\text{kW}, \lambda_1=0.6 \Rightarrow Q_1 = P \times \frac{0.8}{0.6} = \frac{80}{3} \text{ kVar}$$

$$\bar{S} = 20 + j\frac{80}{3} \text{ (kV}\cdot\text{A)}$$

$$(2) \lambda_2=0.9, P \text{ 不变}, Q_2 = P \times \frac{\sqrt{1-\lambda_2^2}}{\lambda_2} = 20 \times \frac{\sqrt{1-(0.9)^2}}{0.9} = 9.69$$

$$\bar{S} = 20 + j9.69 \text{ (kV}\cdot\text{A)}$$

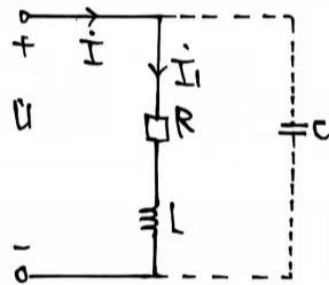
$$\text{故 } Q_C = (9.69 - \frac{80}{3}) \text{ kVA} = -U^2 \cdot WC, W = 100\pi.$$

$$(9.69 - \frac{80}{3}) \times 1000 = -380 \times 380 \times 100\pi \times C, \text{ 解得 } C = 3.74 \times 10^{-4} \text{ F}$$

④ 已知 $f=50\text{Hz}$, $U=220\text{V}$, $P=10\text{kW}$, $\cos\varphi_1=0.6$

要使功率因数提高到 0.9, 问并联多大的电容 C?

另问并联前后的总电流各多大?



$$(1) \text{无 } C \text{ 时}, P=10\text{kW} = UI \cos\varphi_1 \rightarrow \cos\varphi_1=0.6$$

$$Q = P \tan\varphi_1 = 10\text{kW} \times \frac{4}{3} = \frac{40}{3} \text{ kVar} \quad \sin\varphi_1=0.8$$

$$\bar{S}_1 = 10 + j\frac{40}{3} = (10 + j\frac{40}{3}) \text{ kV}\cdot\text{A}$$

$$I_{\text{无 } C \text{ 时}} = \frac{P}{U \cos\varphi_1} = \frac{10\text{kW}}{220 \times 0.6} = \frac{10^4}{220 \times 0.6} = 75.76 \text{ A}$$

2) 有 C 时: P 不变, 因为 U 不变, 流过 R 的 I 不变, $P=I_1^2 R$

$$\lambda_2 = \cos\varphi_2 = 0.9 \quad Q_2 = 10 \cdot \tan\varphi_2 = 10 \times \frac{\sqrt{1-(0.9)^2}}{0.9} = 4.84 \text{ kVar}$$

$$\bar{S}_2 = 10 + jQ_2 = (10 + j4.84) \text{ kV}\cdot\text{A}$$

$$\text{有 } C \text{ 时}, I = \frac{P}{U \cos\varphi_2} = \frac{10^4}{220 \times 0.9} = 50.51 \text{ A}$$

$$Q_C = -\frac{U^2}{WC} = -220 \times 220 \times 100\pi \times C = (4.84 - \frac{40}{3}) \times 10^3$$

$$C = 5.6 \times 10^{-4} \text{ F}$$